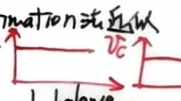
Principle of steady-state converter analysis

STEP 1 drawing ON/OFF 等效 ckt

STEP 2 用 KVL/KCL 在所有元件得方程式

STEP 3 用 small-ripple approximation 或近似 $\Delta i_L \to 0,\ \Delta v_c \to 0$



STEP 4 在电感用 Volt-second balance ⇒ 求电压增益

电容用 amp-second balance 电容器电流闭合

$X_L \to \infty$ $X_C \to \infty$

STEP 5 求 Δi_L 用电感跨压及持续时间

Δv_c 用电容的流入/出及持续时间

STEP 6 求电感电流 $I + \Delta i_L$ 可求电感峰值电流

STEP 0. ① converter 以 operated at steady-state

② The inductor current keep same direction, and continuous

③ the capacitor quite large, $V_o \doteq$ constant

$\Delta v_o \ll V_o$ $V_o = 3.3V$, $\Delta v_o = 10mV$

④ Switching period is T

switch is closed for time DT

open for time (1-D)T

D'T

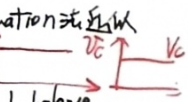
⑤ component are ideal. $D + D' = 1$.

Principle of steady-state converter analysis

STEP 1 drawing ON/OFF 等效 ckt

STEP 2 用 KVL/KCL 在所有元件得方程式

STEP 3 用 small-ripple approximation 或近似 $\Delta i_L \to 0,\ \Delta v_c \to 0$



STEP 4 在电感用 Volt-second balance ⇒ 求电压增益

电容用 amp-second balance 电容器电流闭合

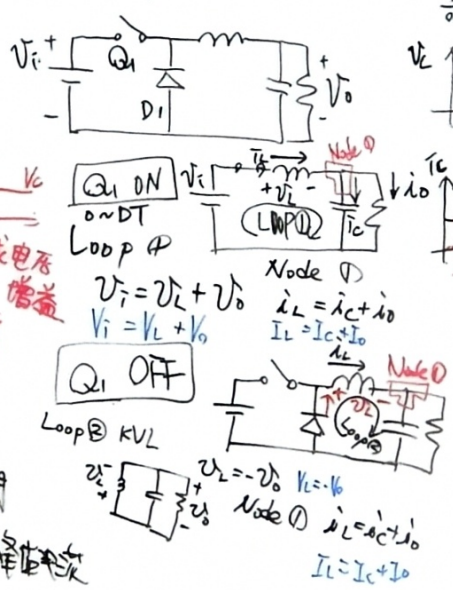
$X_L \to \infty$ $X_C \to \infty$

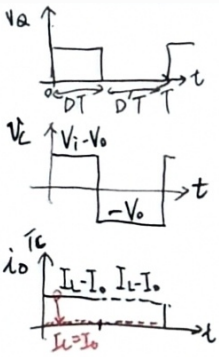
STEP 5 求 Δi_L 用电感跨压及持续时间

Δv_c 用电容的流入/出及持续时间

STEP 6 求电感电流 $I + \Delta i_L$ 可求电感峰值电流

Analysis Buck.





Amp-sec 平衡. $I_L = I_o$

$\langle i_C \rangle_T = 0$

$$\frac{\int_0^{DT}(I_L - I_o)dt + \int_{DT}^{T}(I_L - I_o)dt}{T} = 0$$

$I_L - I_o = 0$. $I_L = I_o$

I_o 就为电流

STEP 4

$\langle v_L \rangle_T = 0 \Rightarrow$

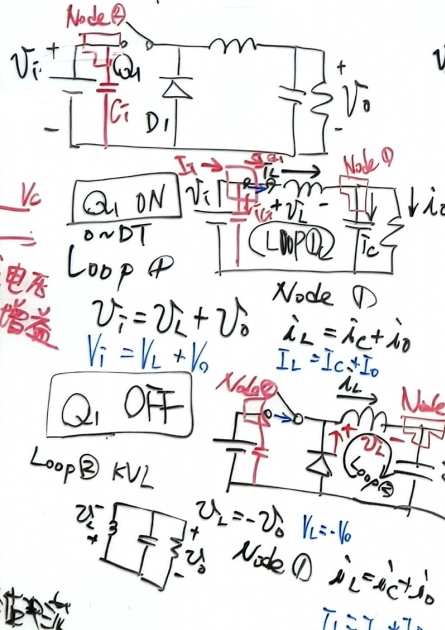
$M(D) = \frac{V_o}{V_i} = D$

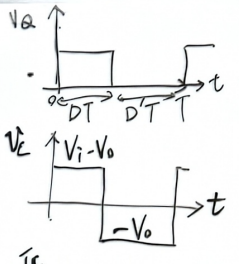
$$\frac{\int_0^{DT} V_i - V_o\,dt + \int_{DT}^{T} -V_o\,dt}{T} = 0$$

$$\frac{(V_i - V_o)(DT - 0) + (-V_o)(T - DT) = 0}{T}$$

$(V_i - V_o)D - V_o D' = 0$, $V_i D = V_o$, $M(D) = \frac{V_o}{V_i} = D$

Analysis Buck.





A.s balance $I_L = I_o$.



pulse current.

Amp-second balance on C_i

ON: $i_i = i_{C_i} + i_{Q_1} \Rightarrow I_i = I_{C_i} + I_{Q_1}$

$I_{Q_1} = I_L$

$I_{C_i} = I_i - I_L$

OFF: $i_{Q_1} = 0$

$i_i = i_{C_i} \Rightarrow I_i = I_{C_i}$

$$\langle i_{C_i} \rangle_T = \frac{(I_i - I_L)DT + I_i D'T}{T} = 0$$

$I_i - I_L D = 0$, $I_i = I_L D$

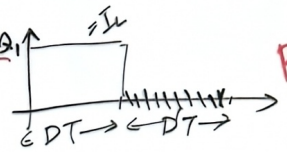
STEP 4

$\langle v_L \rangle_T = 0 \Rightarrow$

$M(D) = \frac{V_o}{V_i} = D$

$I_o = ? = \frac{V_o}{R}$

A·S balance $I_L=I_o$. $I_i=DI_L=DI_o$



pulse current.

$$V_L=L\frac{di_L}{dt} \quad \Delta i_L=\frac{1}{L}\int_0^{DT} v_L\,dt$$

$$\Delta i_L^+=\frac{1}{L}(V_i-V_o)t\Big|_0^{DT}$$

$$=\frac{1}{L}(V_i-V_o)DT>0$$

電感漲相 $\Delta i_L^-=\frac{1}{L}(-V_o)D'T<0.$



伏秒平衡 on L

$$\langle v_L\rangle_T=\frac{\int_0^{DT}v_i\,dt+\int_{DT}^{T}(v_i-v_o)\,dt}{T}=0$$

$$\frac{V_iDT+(V_i-V_o)D'T}{T}=0$$

$$V_i+(-V_o)D'=0 \quad V_i=V_oD' \quad \frac{V_o}{V_i}=\frac{1}{D'}$$

$$\frac{1}{1-D}$$

安秒平衡 on C

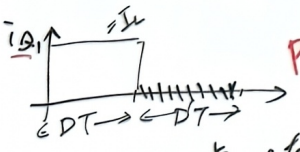
$$\langle i_C\rangle_T=0$$

$$\frac{\int_0^{DT}-I_o\,dt+\int_{DT}^{T}(I_L-I_o)\,dt}{T}=0$$

I_L

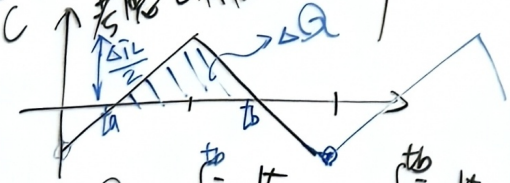
$$\frac{-I_oDT+(I_L-I_o)D'T}{T}=0 \quad \boxed{ID'=I_o}$$

A·S balance $I_L=I_o$. $I_i=DI_L=DI_o$



pulse current

i_C 考慮C有限值



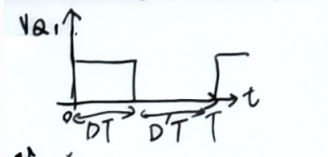
$$C=\frac{\Delta Q}{\Delta V_C}=\frac{\int_{t_a}^{t_b}i_C\,dt}{\Delta V_C} \quad \Delta V_C=\frac{\int_{t_a}^{t_b}i_C\,dt}{C}$$

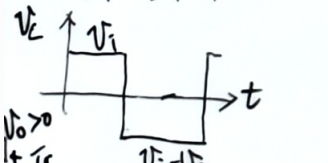
$$t_b-t_a=\frac{I}{2}, \quad \Delta V_C=\frac{\frac{1}{2L}(V_i-V_o)DT\times\frac{I}{2}}{2C}$$

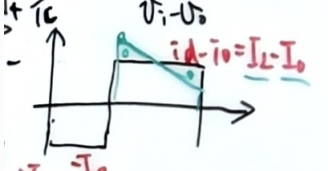
$$=\frac{(V_i-V_o)DT^2}{8LC}$$

$$=\frac{(V_i-V_o)D}{8LCf^2}$$









$$\Delta i_L=\frac{1}{L}\int v_i\,dt=\frac{1}{L}V_iDT$$

$$=\left|\frac{1}{L}(V_i-V_o)D'T\right| \quad <0$$

$$C=\frac{Q}{V}=\frac{\Delta Q}{\Delta V} \quad \Delta V=\frac{\Delta Q}{C}$$

$$\Delta Q^-=-I_oDT \quad \text{代入}$$

$$\Delta Q^+=(I_L-I_o)D'T$$

$$\Delta V=\frac{(I_L-I_o)D'T}{C}=\frac{-I_oDT}{C}$$

STEP 4

$$M(D)=\frac{V_o}{V_i}=\frac{1}{D'}$$

$$I_o=I_LD'$$

$$I_o=?=\frac{V_o}{R}$$

$+i_o$ $\Delta i_L=$

